

# SB-GRPO: Semantic-Balanced GRPO for Enhanced LLM Reasoning

Bang Dinh Nguyen<sup>1,†</sup>, Bao Dang Tran<sup>1,†</sup>, and Minh Quoc Nguyen<sup>1[✉]</sup>

Faculty of Computer Science and Engineering,  
Ho Chi Minh City University of Technology, Vietnam National University,  
Ho Chi Minh City, Vietnam (HCMUT-VNU HCM)  
{bang.nguyendinh, bao.trandang, minhnguyen}@hcmut.edu.vn

**Abstract.** Reinforcement Learning (RL), specifically Group Relative Policy Optimization (GRPO), has emerged as a transformative technique for unlocking deep reasoning in Large Language Models (LLMs). By incentivizing multi-step Chain-of-Thought (CoT) behaviors, GRPO allows models to transcend simple pattern matching. However, current implementations are plagued by critical bottlenecks: advantage vanishing in uniform groups, systematic correction bias during reflection, and inefficient utilization of noisy, high-entropy error samples. We propose SB-GRPO (Semantic-Balanced GRPO), an advanced framework that integrates dynamic multi-layer reflection with semantic error profiling. Our architecture introduces a differentiable Heuristic Confidence Gate that smoothly transitions between standard and negative-enhanced advantage calculation, preventing gradient stalls. We further employ K-Means clustering on reasoning embeddings to identify systematic logical failure modes and construct Static Balanced Batches to stabilize policy gradients. Finally, we derive an information-theoretic loss scaling factor based on the Shannon Entropy of semantic clusters to prioritize fixing systematic errors over random noise. Evaluated under strict parameter-efficient fine-tuning constraints (4-bit QLoRA) using a 7B model, SB-GRPO achieves a highly competitive pass@1 accuracy of 74.80% on the MATH-500 dataset, yielding an absolute improvement of 26.40% over vanilla GRPO. Compared to M-GRPO, a state-of-the-art multi-layer baseline, our method delivers a 2.20% overall gain, with pronounced improvements in complex domains such as Precalculus (+5.68%) and Intermediate Algebra (+4.88%). Furthermore, on the 30-problem AIME 2024 benchmark, it improves pass@1 accuracy from 3.33% (1/30) to 10.00% (3/30).

**Keywords:** Reinforcement Learning · Large Language Models · Mathematical Reasoning · Semantic Entropy

---

<sup>†</sup> : These authors contributed equally to this work.

[✉] : Corresponding author.

## 1 Introduction

The paradigm of Large Language Model (LLM) training has shifted from Supervised Fine-Tuning (SFT) [16] to Reinforcement Learning (RL) [8] alignment. While models such as GPT-4 [1] and Claude [2] demonstrate strong reasoning, reliance on dedicated reward models imposes computational bottlenecks. Group Relative Policy Optimization (GRPO) [10,21] eliminates the critic network by normalizing rewards across rollout groups, enabling policies to optimize reasoning paths without separate reward models.

However, on complex multi-step reasoning, vanilla GRPO faces three critical limitations: (1) **Zero Advantage:** On difficult or saturated prompts, rollouts often share identical outcomes. Reward variance vanishes ( $\sigma \rightarrow 0$ ), driving advantage values to zero [23] and stalling updates. (2) **The Compulsive Correction Fallacy:** Multi-layer [7] and iterative reflection [14] induce overcorrection, biasing models to alter valid steps and causing high false-negative drift. (3) **The Noise-Signal Dichotomy:** Standard RL updates treat all failures equally, failing to distinguish stochastic calculation slips from systematic logical flaws.

To overcome these challenges, we propose Semantic-Balanced GRPO (SB-GRPO). We introduce a differentiable Heuristic Confidence Gate to activate Advantage Calibration [15] with virtual rewards for uniform batches, sustaining gradient flow. For mixed batches, Semantic Error Profiling (SEP) clusters reasoning paths with frozen MiniLM embeddings to isolate systematic errors, forming Static Balanced Batches modulated by an entropy-aware loss scaling factor.

## 2 Related Work

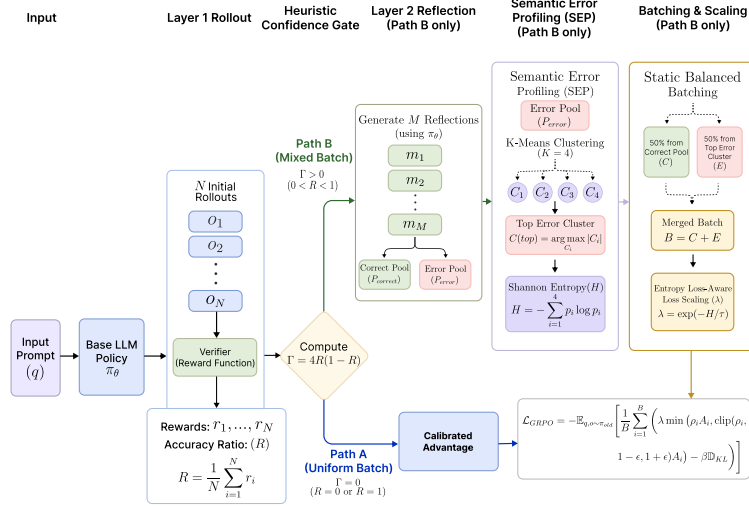
Reinforcement Learning from Human Feedback (RLHF) [4] traditionally relies on Proximal Policy Optimization (PPO) [20], which suffers from heavy computational overhead due to separate reward models. Group Relative Policy Optimization (GRPO) [21] circumvents this bottleneck by substituting the critic with a baseline derived from the average rewards of  $G$  distinct rollouts. This memory-efficient paradigm was further refined by the Advantage Calibration mechanism introduced in NGRPO [15], which incorporates virtual maximum rewards to maintain continuous gradient flow in extreme edge cases. Concurrently, augmenting LLM reasoning via Chain-of-Thought [22] and autonomous self-correction [14] has gained significant traction. Building on this, Multi-Layer GRPO (M-GRPO) [7] employs a two-layer pipeline that utilizes initial outputs as context to verify and correct errors. However, such unconstrained self-reflective frameworks often induce a pronounced correction bias, coercing models to arbitrarily alter valid paths and thereby degrading inference stability.

To mitigate reasoning instability and overcorrection, recent studies focus on uncertainty quantification and structured data curation. Semantic Entropy (SE) [13] provides an information-theoretic metric to distinguish genuine model confusion from generative diversity—a concept SEED-GRPO [3] leverages to dynamically dampen gradient updates for highly uncertain prompts. Additionally,

theoretical insights from Balanced Sampling in DIVA-GRPO [9] emphasize that maintaining an equitable distribution of correct and hard-negative samples is pivotal for stabilizing convergence in sparse-reward environments. Rather than viewing these as isolated techniques, SB-GRPO unifies and adapts these foundational concepts into an active, closed-loop multi-layer framework: dynamic gating resolves the zero-advantage bottleneck, semantic clustering extracts actionable systematic errors, and static balanced batching enforces equitable gradient signals while mitigating correction bias.

### 3 Methodology

This section details the SB-GRPO architecture (Figure 1). To resolve GRPO instability, we introduce a Heuristic Confidence Gate (Sec. 3.2) that dynamically routes uniform batches to use Advantage Calibration, preventing zero advantage. For mixed batches, Layered Generation and Reflection (Sec. 3.3) and Semantic Error Profiling (Sec. 3.4) cluster reasoning trajectories to target systematic failures. The gradients are modulated via an Information-Theoretic Loss Scaling factor (Sec. 3.5) based on error entropy. The step-by-step integration of these modules into a cohesive training iteration is formalized in Algorithm 1.



**Fig. 1.** The complete system architecture of SB-GRPO. The Heuristic Confidence Gate ( $\Gamma$ ) dynamically routes uniform batches to the Advantage Calibration module (Path A) to prevent advantage vanishing. For mixed batches (Path B), the framework applies Multi-Layer Reflection and Semantic Error Profiling (SEP) to extract the Top Error Cluster ( $C_{\text{top}}$ ), forming a Static Balanced Batch optimized by an Entropy-Aware Loss Scaling factor ( $\lambda$ ).

### 3.1 Formal Derivation of SB-GRPO

Let  $q$  be a prompt and  $O = \{o_1, \dots, o_N\}$  be  $N$  rollouts from policy  $\pi_\theta$ . The standard reward for  $o_i$  is  $r_i \in \{0, 1\}$ . In SB-GRPO, we define the accuracy ratio:

$$R = \frac{1}{N} \sum_{i=1}^N r_i$$

### 3.2 Heuristic Confidence Gate

To maintain gradient continuity, we define a differentiable heuristic gating function  $\Gamma(R)$  using the quadratic form:

$$\Gamma(R) = 4R(1 - R) \quad (1)$$

Note that  $\Gamma \in [0, 1]$ , reaching 0 at  $R = 0$  (all wrong) or  $R = 1$  (all correct), and 1 at  $R = 0.5$ . This parabolic form ensures that purely correct or incorrect batches seamlessly switch to Calibrated Advantage loss, while balanced batches optimally utilize multi-layer reflection without gradient spikes. This gate selects the advantage calculation method:

$$A_i = \Gamma \cdot A_{\text{std},i} + (1 - \Gamma) \cdot A_{\text{calib},i} \quad (2)$$

where  $A_{\text{calib}}$  uses a virtual reward  $r_{\text{virtual}} = 1$  to inject a negative signal even when all  $r_i = 0$ . Injecting this virtual reward increases the mean  $\mu$ , which in turn pushes the  $A_{\text{std},i} = (r_i - \mu)/\sigma$  of incorrect samples into explicitly negative values rather than leaving them stuck at 0. The standard advantage is computed as  $A_{\text{std},i} = (r_i - \mu)/\sigma$ , where  $\mu$  and  $\sigma$  are the mean and standard deviation of rewards within the group. When variance approaches zero,  $A_{\text{std}}$  approaches zero, making  $A_{\text{calib}}$  critical for continuous training. Conversely, when  $R = 1$  (all correct), injecting  $r_{\text{virtual}} = 1$  maintains a zero advantage, correctly signaling that no policy update is required for an already mastered prompt.

To illustrate this adaptive mechanism for sample size  $N = 8$ : (1) **Case 1 (All-Incorrect,  $R = 0$ ):**  $\Gamma = 0$ . Standard GRPO yields  $A_{\text{std}} = 0$ , halting learning. Advantage Calibration injects  $r_{\text{virtual}} = 1$  ( $\mu' \approx 0.11$ ,  $\sigma' \approx 0.31$ ), assigning  $A \approx -0.35$  to incorrect samples to drive exploration. (2) **Case 2 (Low-Accuracy Mixed, 1 correct,  $R = 0.125$ ):**  $\Gamma = 0.4375$ . Standard GRPO overweights the lucky guess ( $A_{\text{std}} \approx +2.65$ ) and weakly penalizes wrong ones ( $-0.38$ ). Routed advantage rebalances to  $A_{\text{correct}} \approx +2.21$  and  $A_{\text{wrong}} \approx -0.47$ , mitigating lucky guess variance. (3) **Case 3 (High-Accuracy Mixed, 7 correct,  $R = 0.875$ ):**  $\Gamma = 0.4375$ . Standard GRPO computes  $+0.38$  and  $-2.65$ . Routed advantage minimally shifts these to  $+0.36$  and  $-2.75$ , stably favoring exploitation while strictly penalizing rare failures.

**Algorithm 1** SB-GRPO Training Step

---

```

1: Input: Prompt  $q$ , Policy  $\pi_\theta$ , Verifier  $V$ 
2: Generate  $N$  rollouts  $O = \{o_1, \dots, o_N\}$  via vLLM;  $R \leftarrow \text{Accuracy}(O, V)$ ;  $\Gamma \leftarrow 4R(1 - R)$ 
3: if  $\Gamma > 0$  then
4:   Generate  $M$  reflections per rollout  $\rightarrow$  Pool  $P$ ; Cluster  $P_{\text{error}}$  into  $\{C_1 \dots C_K\}$ 
5:    $\lambda \leftarrow \exp(-\text{Entropy}(C)/\tau)$ ;  $B \leftarrow$  Sample  $S/2$  from  $P_{\text{correct}}$  and  $S/2$  from  $C_{\text{top}}$ 
6: else
7:    $\lambda \leftarrow 1.0$ ;  $B \leftarrow$  Sample  $S$  uniformly from  $O$ ; Use Calibrated Advantage  $A_{\text{calib}}$ 
8: end if
9: Training Phase: Flush vLLM cache  $\rightarrow$  Init PyTorch Trainer
10:  $\rho \leftarrow \frac{\pi_\theta}{\pi_{\text{old}}}$ ;  $\mathcal{L}_{\text{surv}} \leftarrow \min(\rho A, \text{clip}(\rho, 1 - \epsilon, 1 + \epsilon) A)$ 
11:  $\mathcal{L}_{\text{GRPO}} \leftarrow \frac{1}{|B|} \sum (-\lambda \mathcal{L}_{\text{surv}} + \beta \mathbb{D}_{KL})$ ; Backward and Optimizer Step

```

---

**3.3 Layered Generation and Reflection**

Rather than relying on a single-pass generation, SB-GRPO isolates the exploration and verification phases into a two-layer pipeline. This deliberate separation allows the model to systematically audit its own reasoning paths before gradient updates are computed. The process is defined as follows:

**Layer 1 (Rollout):**  $N = 8$  samples are generated with temperature  $\tau = 0.7$  to explore the solution space.

**Layer 2 (Reflection):** If  $\Gamma > 0$ , the model performs  $M = 2$  reflections per rollout using a neutral system prompt: *"Review the solution logic. Only correct it if errors are found; otherwise, maintain consistency."* This creates a pool of  $N \times M = 16$  reasoning paths.

**3.4 Semantic Error Profiling (SEP)**

To effectively disambiguate structural reasoning flaws from superficial generation noise, the SEP module projects all incorrect trajectories into a dense semantic space. Specifically, incorrect samples are vectorized utilizing a frozen *all-MiniLM-L6-v2* encoder [17]. Subsequently, K-Means clustering [19] (empirically set to  $K = 4$ ) is applied to partition these latent embeddings into a discrete set of semantic clusters  $C = \{C_1, \dots, C_K\}$ . By identifying the top error cluster, defined mathematically as  $C_{\text{top}} = \arg \max_k |C_k|$ , we isolate the most prevalent logical failure mode associated with the given prompt. This isolation provides a concentrated learning signal for targeted error correction, ensuring the policy's optimization against systematic misconceptions rather than stochastic, trivial errors.

**3.5 Information-Theoretic Loss Scaling**

To mathematically quantify the epistemic uncertainty inherent in the model's failure modes, we compute the Shannon Entropy  $H$  over the semantic cluster

distribution. Let  $p_k = |C_k| / \sum_{j=1}^K |C_j|$  denote the probability mass of the  $k$ -th cluster. The entropy is defined as:

$$H = - \sum_{k=1}^K p_k \log(p_k) \quad (3)$$

Based on this uncertainty metric, we introduce a dynamic loss scaling factor  $\lambda = \exp(-H/\tau)$ , where  $\tau$  serves as a temperature hyperparameter (set empirically to  $\tau = 1.0$ ). This scaling factor is directly applied to modulate the policy gradient. Intuitively, a high entropy value signifies a highly dispersed, random distribution of superficial errors (e.g., stochastic arithmetic mistakes), thereby dampening the gradient update to prevent the model from overfitting to noise. Conversely, a low entropy value indicates a concentrated, systematic logical flaw, triggering a remarkably larger, more aggressive optimization step to rectify the foundational misunderstanding.

### 3.6 Semantic-Balanced GRPO Objective Function

The final optimization objective integrates the standard PPO clipped surrogate loss, the KL divergence penalty, and our novel entropy-aware loss scaling factor  $\lambda$ . For a prompt  $q$  and a static balanced batch  $B$  of size  $S$ , the policy  $\pi_\theta$  is updated by maximizing the objective:

$$\mathcal{J}_{\text{SB-GRPO}}(\theta) = \frac{1}{S} \sum_{i=1}^S \left[ \lambda \min \left( \frac{\pi_\theta(o_i|q)}{\pi_{\text{old}}(o_i|q)} A_i, \text{clip} \left( \frac{\pi_\theta(o_i|q)}{\pi_{\text{old}}(o_i|q)}, 1 - \epsilon, 1 + \epsilon \right) A_i \right) - \beta \mathbb{D}_{\text{KL}}(\pi_\theta \| \pi_{\text{ref}}) \right] \quad (4)$$

where  $A_i$  is the dynamically routed advantage (via the Heuristic Confidence Gate  $\Gamma$ ),  $\epsilon$  is the clipping hyperparameter, and  $\beta$  controls the KL penalty against the reference model  $\pi_{\text{ref}}$ . By scaling the surrogate objective with  $\lambda$ , the magnitude of the policy update is strictly regulated by the semantic concentration of the error clusters, forcing the model to heavily penalize and unlearn systematic logical flaws while conservatively adjusting for noisy, random errors.

## 4 Datasets and Experimental Setup

### 4.1 Datasets

We utilize well-established reasoning datasets for training and evaluation. **Training Data:** We aggregate **MATH (Levels 3–5)** [11] (5,586 multi-step problems filtered to exclude trivial tiers) and **GSM8K** [5] (7,473 training problems). **Evaluation Benchmarks:** We evaluate on **MATH-500** (500 curated test problems), **AIME 2024 & 2025** (30 elite problems each), **GPQA Diamond** [18] (198 PhD-level science questions), and **GSM8K** (1,319 test questions).

## 4.2 Experimental Setup

All experiments, including our SB-GRPO and baselines (Vanilla GRPO, M-GRPO), are conducted under a strictly controlled environment. We adopt `Qwen/Qwen2.5-7B-Instruct` as the foundational model. Under memory constraints of running a 7B model and generation engine simultaneously, we apply 4-bit QLoRA [12,6] (`load_in_4bit=True`, `use_peft=True`) with DeepSpeed ZeRO-3 and Flash Attention 2 on a single NVIDIA H100 GPU (80GB VRAM). Models are trained for 1 epoch with learning rate  $1.0 \times 10^{-5}$ , max prompt length 2048, and completion length 1024, generating  $N = 8$  completions per prompt.

**Evaluation Protocol:** To measure both peak deterministic performance and variance on small discrete test suites (e.g., AIME’s 30 problems), we report both single-sample greedy decoding and multi-pass evaluation (Table 1) averaged over 4 independent generation passes with distinct seeds on the trained checkpoint.

**Hardware-Aware Static Tensor Shaping:** To maximize GPU throughput on the NVIDIA H100 and completely mitigate memory fragmentation induced by variable sequence lengths, we strictly avoid dynamic padding. Building upon the variance reduction principles of DIVA-GRPO, we construct a Static Balanced Batch of a fixed size  $S = 16$ . Specifically, we sample  $S/2$  trajectories from the Correct Pool and the remaining  $S/2$  from the Top Error Cluster ( $C_{\text{top}}$ ). By enforcing Sampling with Replacement, we guarantee that the tensor shape remains entirely constant across all training steps. This static geometric configuration minimizes memory allocation overhead, thereby fully maximizing the computational efficiency of the DeepSpeed ZeRO-3 backend under constrained VRAM environments.

## 5 Experimental Results

### 5.1 Performance Evaluation

All models were evaluated using the `lighteval` framework on the exact same hardware configuration to ensure fairness. Table 1 compares SB-GRPO with vanilla GRPO (which lacks reflection entirely) and M-GRPO [7]. M-GRPO is chosen as our primary baseline because it represents the state-of-the-art in multi-layer reinforcement learning for self-correction. By utilizing the same two-layer reflection structure but sampling revisions uniformly without semantic clustering or balancing, M-GRPO serves as a direct control group (ablation). This allows us to isolate the performance gains driven specifically by our semantic profiling and balanced batching mechanisms, minimizing architectural confounding variables.

### 5.2 Analysis

The results demonstrate that SB-GRPO’s core strength lies in resolving high-complexity reasoning tasks:

**Table 1.** Pass@1 Accuracy (%) and absolute gains ( $\Delta$ ) on Advanced Reasoning Benchmarks under 1-Sample Greedy Decoding and Multi-Sample (4-Pass) Evaluation (mean  $\pm$  stderr from `lighteval`).  $\Delta_{S-V}$  and  $\Delta_{S-M}$  denote absolute improvements over Vanilla GRPO and M-GRPO.

| Dataset                                                               | Vanilla          | M-GRPO           | <b>SB-GRPO</b>                     | $\Delta_{S-V}$ | $\Delta_{S-M}$ |
|-----------------------------------------------------------------------|------------------|------------------|------------------------------------|----------------|----------------|
| <i>Single-Sample Greedy Decoding (1-Pass)</i>                         |                  |                  |                                    |                |                |
| MATH-500                                                              | 48.40            | 72.60            | <b>74.80</b>                       | <b>+26.40</b>  | <b>+2.20</b>   |
| AIME 2024                                                             | 3.33             | 6.67             | <b>10.00</b>                       | <b>+6.67</b>   | <b>+3.33</b>   |
| AIME 2025                                                             | 3.33             | <b>6.67</b>      | <b>6.67</b>                        | <b>+3.34</b>   | 0.00           |
| GPQA Diamond                                                          | 30.30            | 31.82            | <b>32.83</b>                       | <b>+2.53</b>   | <b>+1.01</b>   |
| GSM8K                                                                 | 56.33            | <b>70.51</b>     | 69.22                              | <b>+12.89</b>  | -1.29          |
| <i>Multi-Sample Evaluation (4-Pass, mean <math>\pm</math> stderr)</i> |                  |                  |                                    |                |                |
| MATH-500                                                              | 47.60 $\pm$ 1.85 | 72.30 $\pm$ 1.68 | <b>75.65 <math>\pm</math> 1.65</b> | <b>+28.05</b>  | <b>+3.35</b>   |
| AIME 2024                                                             | 1.67 $\pm$ 1.67  | 10.83 $\pm$ 4.59 | <b>12.50 <math>\pm</math> 4.91</b> | <b>+10.83</b>  | <b>+1.67</b>   |
| AIME 2025                                                             | 0.83 $\pm$ 0.83  | 6.67 $\pm$ 3.38  | <b>8.33 <math>\pm</math> 3.46</b>  | <b>+7.50</b>   | <b>+1.66</b>   |
| GPQA Diamond                                                          | 28.91 $\pm$ 2.17 | 29.42 $\pm$ 2.36 | <b>33.08 <math>\pm</math> 2.45</b> | <b>+4.17</b>   | <b>+3.66</b>   |

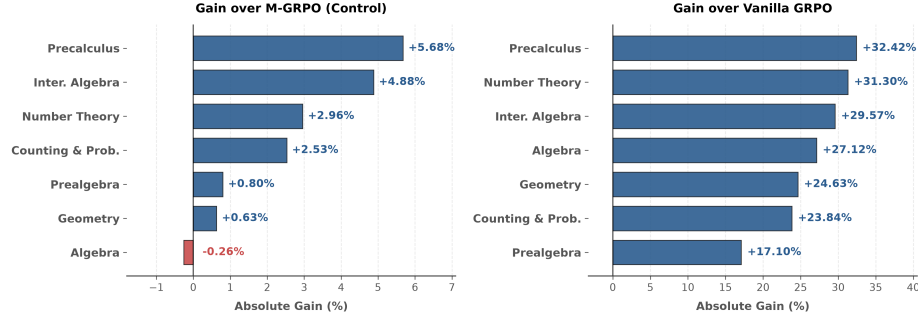
**Single-Sample Evaluation (Table 1, Top):** Under 1-sample greedy decoding, SB-GRPO achieves +2.20% on MATH-500, +3.33% on AIME 2024, and +1.01% on GPQA Diamond over M-GRPO. On AIME 2025, SB-GRPO achieves 6.67%, performing on par with M-GRPO ( $\Delta_{S-M} = 0.00\%$ ) in this single-sample snapshot. Because AIME comprises only 30 problems, single-pass evaluations exhibit discrete quantization granularity ( $\pm 3.33\%$  per problem), where a single question dictates the metric. This directly motivated our multi-pass protocol.

**Multi-Sample Evaluation (Table 1):** Across 4 generation passes with distinct seeds on the trained policy, the advantages of semantic balancing become statistically robust. Table 1 reports Pass@1 accuracy alongside standard errors (mean  $\pm$  stderr) directly computed by `lighteval`. SB-GRPO achieves  $75.65 \pm 1.65\%$  on MATH-500 (+3.35% over M-GRPO) and  $33.08 \pm 2.45\%$  on GPQA Diamond (+3.66%). Crucially, on AIME 2025, multi-pass evaluation breaks the single-sample tie, with SB-GRPO reaching  $8.33 \pm 3.46\%$  (10/120 solved) vs.  $6.67 \pm 3.38\%$  (8/120 solved) for M-GRPO. On AIME 2024, SB-GRPO achieves  $12.50 \pm 4.91\%$  (15/120) compared to  $10.83 \pm 4.59\%$  (13/120) for M-GRPO and  $1.67 \pm 1.67\%$  (2/120) for Vanilla GRPO, confirming consistent multi-seed gains.

**Sub-category Analysis (Table 2):** As depicted in Fig. 2, SB-GRPO exhibits its most substantial advantages in rigorous multi-stage disciplines: Precalculus (+5.68%) and Intermediate Algebra (+4.88%). For saturated domains like elementary Algebra ( $\sim 89\%$  accuracy), failure modes are predominantly sporadic arithmetic slips rather than structural logical flaws; the entropy loss scaling factor adaptively downscales updates to preserve policy representations, yielding performance parity (89.13% vs 89.39%, -0.26%).

**Table 2.** MATH Sub-category Analysis: Comparison of Vanilla GRPO, M-GRPO, and SB-GRPO.  $\Delta_{S-V}$  and  $\Delta_{S-M}$  denote the absolute improvements of SB-GRPO over Vanilla GRPO and M-GRPO, respectively.

| Category               | Accuracy (%) |              |                | Absolute Gains (%) |                |
|------------------------|--------------|--------------|----------------|--------------------|----------------|
|                        | Vanilla      | M-GRPO       | <b>SB-GRPO</b> | $\Delta_{S-V}$     | $\Delta_{S-M}$ |
| Algebra                | 62.01        | <b>89.39</b> | 89.13          | <b>+27.12</b>      | -0.26          |
| Counting & Probability | 43.25        | 64.56        | <b>67.09</b>   | <b>+23.84</b>      | <b>+2.53</b>   |
| Geometry               | 32.99        | 56.99        | <b>57.62</b>   | <b>+24.63</b>      | <b>+0.63</b>   |
| Intermediate Algebra   | 25.14        | 49.83        | <b>54.71</b>   | <b>+29.57</b>      | <b>+4.88</b>   |
| Number Theory          | 42.22        | 70.56        | <b>73.52</b>   | <b>+31.30</b>      | <b>+2.96</b>   |
| Prealgebra             | 65.33        | 81.63        | <b>82.43</b>   | <b>+17.10</b>      | <b>+0.80</b>   |
| Precalculus            | 24.54        | 51.28        | <b>56.96</b>   | <b>+32.42</b>      | <b>+5.68</b>   |
| <b>Overall</b>         | 42.21        | 66.32        | <b>68.78</b>   | <b>+26.57</b>      | <b>+2.46</b>   |



**Fig. 2.** Absolute performance gains of SB-GRPO on MATH sub-categories. **Left:** Compared to M-GRPO, showing pronounced improvements in *Intermediate Algebra* (+4.88%) and *Precalculus* (+5.68%). **Right:** Compared to Vanilla GRPO, highlighting broad gains achieved by multi-layer reasoning.

### 5.3 Ablation Analysis: Deconstructing Performance Gains

We deconstruct the core mechanisms of SB-GRPO to examine their operational effects:

**1. Heuristic Confidence Gate ( $\Gamma$ ) & Advantage Calibration:** In standard GRPO, uniform rollouts ( $R = 0$  or  $1$ ) cause reward variance to vanish ( $\sigma \rightarrow 0$ ), driving  $A_{\text{std}} = 0$  and stalling updates (Sec. 3.2). The gate smoothly routes uniform batches to virtual-reward calibration ( $A_{\text{calib}}$ ), sustaining continuous gradient flow without optimization spikes.

**2. Semantic Error Profiling (SEP) & Balanced Batching:** M-GRPO serves as our structural control (ablation), using two-layer reflection with uniform revision sampling (72.60% on MATH-500). Replacing uniform sampling with SEP and static 50/50 balanced batches ( $S = 16$ ) concentrates gradients on the dominant error cluster ( $C_{\text{top}}$ ) while maintaining verified path parity, delivering an immediate +2.20% gain.

**3. Information-Theoretic Loss Scaling ( $\lambda$ ):** Without entropy scaling ( $\lambda = 1.0$ ), the policy treats all errors equally. When errors are stochastic slips (high cluster entropy  $H$ , e.g., GSM8K),  $\lambda = \exp(-H/\tau)$  attenuates updates to prevent overfitting to superficial noise. When errors concentrate into a systematic misconception (low  $H$ ),  $\lambda$  scales up to enforce decisive unlearning.

**4. Structural Mitigation of Correction Bias:** Unguided reflection (M-GRPO) frequently induces models to second-guess valid intermediate steps, flipping correct solutions into incorrect revisions (false-negative drift). SB-GRPO structurally prevents this via: (i) an objective, neutral reflection prompt (Sec. 3.3) that eliminates bias toward unwarranted modifications, and (ii) Static Balanced Batching ( $S/2$  correct,  $S/2$  from  $C_{\text{top}}$ ), ensuring negative paths do not skew updates.

## 5.4 Discussion

**The GSM8K Trade-Off:** SB-GRPO scores 69.22% on GSM8K vs. M-GRPO’s 70.51%. GSM8K consists of elementary word problems where errors stem from sporadic calculation slips rather than structural flaws. Consequently, semantic entropy  $H$  remains high, prompting  $\lambda$  to dampen gradient updates. While M-GRPO overfits to arithmetic noise, SB-GRPO preserves representational stability, driving decisive superiority on advanced benchmarks.

**Qualitative Inspection of Semantic Error Clusters:** Qualitative inspection of  $C_{\text{top}}$  confirms our core premise: on Intermediate Algebra and Precalculus in MATH-500, incorrect rollouts within  $C_{\text{top}}$  consistently share identical mathematical misconceptions—such as misapplying sign reversals in quadratic inequalities or neglecting boundary conditions in logarithmic domains—rather than arbitrary lexical overlap. This demonstrates that frozen MiniLM embeddings reliably isolate systematic logical flaws.

## 6 Conclusion and Future Work

In this paper, we introduced Semantic-Balanced Group Relative Policy Optimization (SB-GRPO), addressing advantage vanishing and correction bias in LLM reasoning. By synthesizing a Heuristic Confidence Gate with Semantic Error Profiling and an Information-Theoretic Loss Scaling factor, SB-GRPO stabilizes gradients across skewed batches while actively targeting systematic logical flaws. Under strict 4-bit QLoRA constraints on a single H100 GPU, SB-GRPO achieves 74.80% on MATH-500 (+2.20% over M-GRPO) and 10.00% on AIME 2024, excelling in multi-stage domains like Precalculus (+5.68%) and Intermediate Algebra (+4.88%). Future work will explore extending this entropy-aware mechanism to multi-modal reasoning and evaluating cross-architecture generalizability beyond Qwen2.5-7B.

## References

1. Achiam, J., Adler, S., Agarwal, S., et al.: Gpt-4 technical report (2023), <https://arxiv.org/abs/2303.08774>
2. Anthropic: The claude 3 model family: Opus, sonnet, haiku (2024), <https://api.semanticscholar.org/CorpusID:268232499>
3. Chen, M., Chen, G., Wang, W., Yang, Y.: SEED-GRPO: Semantic entropy enhanced GRPO for uncertainty-aware policy optimization (2025), <https://arxiv.org/abs/2505.12346>
4. Christiano, P.F., Leike, J., Brown, T.B., Martic, M., Legg, S., Amodei, D.: Deep reinforcement learning from human preferences. In: Proceedings of the 31st International Conference on Neural Information Processing Systems. pp. 4302–4310. NIPS’17 (2017)
5. Cobbe, K., Kosaraju, V., Bavarian, M., Chen, M., et al.: Training verifiers to solve math word problems (2021), <https://arxiv.org/abs/2110.14168>
6. Dettmers, T., Pagnoni, A., Holtzman, A., Zettlemoyer, L.: QLoRA: Efficient fine-tuning of quantized LLMs (2023), <https://arxiv.org/abs/2305.14314>
7. Ding, F., Wang, B., Zeng, Z., Wang, Y.: Multi-layer GRPO: Enhancing reasoning and self-correction in large language models (2025), <https://arxiv.org/abs/2506.04746>
8. François-Lavet, V., Henderson, P., Islam, R., Bellemare, M.G., Pineau, J.: An introduction to deep reinforcement learning. Foundations and Trends in Machine Learning **11**(3-4), 219–354 (2018)
9. Gao, H., Zhang, Z., Pang, L., Guo, F., et al.: DIVA-GRPO: Enhancing multimodal reasoning through difficulty-adaptive variant advantage (2026), <https://arxiv.org/abs/2603.01106>
10. Guo, D., Yang, D., et al.: Deepseek-r1 incentivizes reasoning in LLMs through reinforcement learning. Nature **645**(8081), 633–638 (2025)
11. Hendrycks, D., Burns, C., Kadavath, S., Arora, A., et al.: Measuring mathematical problem solving with the MATH dataset (2021), <https://arxiv.org/abs/2103.03874>
12. Hu, E.J., Shen, Y., Wallis, P., Allen-Zhu, Z., et al.: LoRA: Low-rank adaptation of large language models (2021), <https://arxiv.org/abs/2106.09685>
13. Kuhn, L., Gal, Y., Farquhar, S.: Semantic uncertainty: Linguistic invariances for uncertainty estimation in natural language generation (2023), <https://arxiv.org/abs/2302.09664>
14. Madaan, A., Tandon, N., Gupta, P., Hallinan, S., et al.: Self-refine: Iterative refinement with self-feedback (2023), <https://arxiv.org/abs/2303.17651>
15. Nan, G., Chen, S., Huang, J., Lu, M., et al.: NGRPO: Negative-enhanced group relative policy optimization (2025), <https://arxiv.org/abs/2509.18851>
16. Ouyang, L., Wu, J., Jiang, X., Almeida, D., et al.: Training language models to follow instructions with human feedback (2022), <https://arxiv.org/abs/2203.02155>
17. Reimers, N., Gurevych, I.: Sentence-BERT: Sentence embeddings using siamese BERT-networks (2019), <https://arxiv.org/abs/1908.10084>
18. Rein, D., Hou, B.L., Stickland, A.C., Petty, J., et al.: GPQA: A graduate-level google-proof Q&A benchmark (2023), <https://arxiv.org/abs/2311.12022>
19. Sarasvananda, I.B., Wardoyo, R., Sari, A.: The k-means clustering algorithm with semantic similarity to estimate the cost of hospitalization. IJCCS **13** (2019). <https://doi.org/10.22146/ijccs.45093>

20. Schulman, J., Wolski, F., Dhariwal, P., Radford, A., Klimov, O.: Proximal policy optimization algorithms (2017), <https://arxiv.org/abs/1707.06347>
21. Shao, Z., Wang, P., Zhu, Q., Xu, R., et al.: DeepSeekMath: Pushing the limits of mathematical reasoning in open language models (2024), <https://arxiv.org/abs/2402.03300>
22. Wei, J., Wang, X., Schuurmans, D., Bosma, M., et al.: Chain-of-thought prompting elicits reasoning in large language models. In: Proceedings of the 36th International Conference on Neural Information Processing Systems. NIPS '22 (2022)
23. Zhang, Z., Han, Z., Mavromatis, C., et al.: Train less, learn more: Adaptive efficient rollout optimization for group-based reinforcement learning (2026), <https://arxiv.org/abs/2602.14338>